

Ashutosh Gupta

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EDUCATION

Thapar Institute of Engineering and Technology

Bachelor of Engineering in Computer Engineering

Patiala

2022 – 2026

Cambridge School

Class XII (PCM), CBSE — 81.4%

Rewari

2021

Cambridge School

Class X, CBSE — 82.2%

Rewari

2019

EXPERIENCE

Virves Software Solutions

Remote

AI Trainee

June 2026 – Present

- Developed a FastAPI-based OCR validation backend for Aadhaar, PAN Card, and Driving License documents using EasyOCR, image preprocessing, and rule-based decision logic.
- Built document-number extraction, PAN holder-name extraction, strict profile-name matching, and cross-document mismatch detection from noisy OCR text.
- Improved OCR pipeline reliability and runtime using staged preprocessing, image resizing, candidate deduplication, and early-exit optimization while maintaining a fully passing 139+ unit test suite.

TECHNICAL SKILLS

Languages: Python, C, C++, SQL, R

Computer Vision & OCR: EasyOCR, Image Preprocessing, OCR Pipelines, Document Image Analysis, OpenCV, PyPDF2

AI/ML: Machine Learning, Deep Learning, PyTorch, TensorFlow, Keras, Scikit-learn, Model Evaluation, Feature Engineering

Backend & APIs: FastAPI, Flask, REST APIs, SQLite

Tools & Platforms: NumPy, Pandas, Matplotlib, Seaborn, Git, GitHub, VS Code, Jupyter Notebook, Google Colab, Render

PROJECTS

Resume Screener | Python, Scikit-learn, Gensim, PyPDF2, OpenAI GPT-3.5, Streamlit, Plotly

[Link](#)

- Built a resume-JD matcher using TF-IDF, Word2Vec, and keyword overlap scoring to generate an overall match percentage.
- Processed PDF resumes and raw job descriptions using PyPDF2-based text extraction and cleaning pipeline.
- Integrated GPT-3.5 to generate personalized resume improvement suggestions with an interactive Streamlit dashboard.

Fake News Detector | Python, Scikit-learn, FastAPI, NLTK, TF-IDF, Ensemble ML, Render

[Link](#)

- Trained multiple ML models on 40K+ news articles using TF-IDF and ensemble voting, achieving ~99.3% accuracy.
- Built an NLP pipeline with lemmatization, stopword removal, title weighting, and bigram TF-IDF features.
- Developed and deployed a FastAPI app with prediction, health, and model-info endpoints on Render.

ShieldAI Fraud Detection System | Python, Flask, Scikit-learn, Random Forest, SQLite, JavaScript

[Link](#)

- Trained a Random Forest model on 10,000 transactions with behavioral features, achieving 97%+ accuracy and 0.99 ROC-AUC.
- Handled class imbalance using StandardScaler pipeline and class-weight balancing for fraud detection.
- Built a Flask REST API with SQLite persistence and a JavaScript dashboard for live KPIs and risk alerts.

CERTIFICATIONS

Introduction to Python | Coding Ninjas

[Link](#)

Data Structures & Algorithms in Python | Coding Ninjas

[Link](#)

Natural Language Processing Specialization | Coursera

[Link](#)

20+ Guided Projects | Coursera